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1. What is a multi-factor asset-pricing model? Why were multi-factor models developed and what are their potential benefits and limitations?

2. Give an overview of the CAPM. What is the single factor used to explain the expected return for a security or a portfolio?

1.
A Multi-factor asset pricing model is a model that explains the expected return of a portfolio by its beta to many other(not only one) risk factors, such as interest rate, size value, and momentum. We use a multi-model approach as it provides a more comprehensive review than the CAPM, which focuses solely on one factor. Thus, it helps manage risk more efficiently and explains fluctuations in return more effectively.
Benefits:

- Deliver overall returns
- Manage risk more effectively by considering many factors in model.

Limitations:

- Choosing many factors can lead to overfitting, as these factors also vary over time, making it difficult to keep track.
- Estimation requires more data, which increases complexity and the potential for error.

2. Overview of CAPM:

CAPM is a single-factor model that measures the expected return of a portfolio or securities to the overall market performance.

Formular:
$E(Ri)= Rf+\beta i\ (E(Rm)-Rf)$

Where
$E(Ri)$ = Expected return of the asset i
Rf = Risk-free rate
βi = Beta of the asset
$E(Rm)-Rf$ = Market risk premium

The single factor used to explain the expected return of a security or portfolio is the market risk premium $E(Rm)-Rf$, which measures the systematic risk that investors is compensated.